Topology with declared operators

LLM-assisted specimen. Four foundation results; the research theorem targets remain pending. These foundation results exercise the implemented language. They do not assert the five pending paper theorems.

Result 1 (inverse image intersection). [Topology foundations, inverse image intersection]
Let X and Y be types. Let f be a function from X to Y. Let S and T be subsets of Y. Then

$$f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T).$$

Result 2 (image membership). [Topology foundations, image membership]
Let X and Y be types. Let f be a function from X to Y. Let S be a subset of X. Let y be an element of Y. Then

$$y \in f(S) \text{ if and only if } (\text{there exists } x : X,\ x \in S \text{ and } f(x) = y)\,.$$

Result 3 (continuous composition). [Topology foundations, continuous composition]
Let X, Y and Z be types. Let t be a topology on X. Let u be a topology on Y. Let v be a topology on Z. Let f be a function from X to Y. Let g be a function from Y to Z. Assume f is continuous from t to u (`continuity of f from t to u`). Assume g is continuous from u to v (`continuity of g from u to v`). Then

$$g \circ f \text{ is continuous from } t \text{ to } v.$$

Result 4 (family intersection membership). [Topology foundations, family intersection membership]
Let I and X be types. Let *sets* be a function from I to $\mathcal{P}(X)$. Let x be an element of X. Then

$$x \in \text{intersection family}(\mathit{sets}) \text{ if and only if } (\text{for every } i : I,\ x \in \mathit{sets}(i))\,.$$